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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU BVoc (Honours) Regular Examination April 2026
SECOND SEMESTER
Skill Development Components (SDC) - Human Body: Structure, Function & Energy Metabolism.
(2025 Admission onwards)

Part A: Multiple Choice Questions (MCQs)

Answer key for questions 1 to 12. Each carries 1 mark.

1. Which of the following is a flat bone?

Correct Answer: c). Sternum

2. Dehydration can negatively affect exercise performance by:

Correct Answer: b). Reducing blood volume and thermoregulation

3. Understanding joint kinetics helps coaches and athletes to:

Correct Answer: a). Reduce injury risk and improve performance

4. Understanding kinematic chains helps athletes and coaches to:

Correct Answer: a). Design exercises that enhance performance, efficiency, and injury prevention

5. Understanding anaerobic adaptations helps athletes to:

Correct Answer: a). Improve explosive power, strength, and performance while reducing injury risk

6. Proper warm-up before sports improves joint function by:

Correct Answer: b). Increasing joint lubrication and flexibility

7. Which hormone is primarily responsible for increasing blood glucose levels during exercise?

Correct Answer: b). Glucagon

8. Which of the following is a key application of biomechanics in sports?

Correct Answer: a). Designing effective training techniques

9. Kinesiology is primarily the study of:

Correct Answer: b). Human movement, including muscles, joints, and biomechanics

10. Which cardiovascular adaptation is commonly seen after prolonged aerobic training?

Correct Answer: a). Lower resting heart rate and increased cardiac output

11. The knee-jerk reflex is an example of a:

Correct Answer: b). Monosynaptic reflex

12. The basic structural and functional unit of the human body is the:

Correct Answer: b). Cell

Part B: Short Answer Questions

answering any 5 questions. Each carries 2 marks.

13. Why is knowledge of center of gravity, balance, and stability important for injury prevention in athletes?

Knowing how to control the Center of Gravity (CoG) relative to the base of support helps athletes maintain stability.

It prevents awkward falls, improves landing mechanics, and minimizes unexpected joint shearing forces during rapid directional shifts.

14. How does understanding kinesiology improve performance in lifting, running, and throwing?

Kinesiology provides an understanding of muscle recruitment and joint mechanics.

Athletes can use it to perfect their form (e.g., proper hip hinge in lifting, stride efficiency in running), eliminating wasted energy and maximizing power output.

15. How can understanding fatigue mechanisms help athletes prevent injury during high-intensity training?

Fatigue degrades neural coordination, depletes substrate levels, and reduces muscle force production.

Recognizing these pathways allows athletes and coaches to schedule timely rest periods and avoid structural failure caused by compromised mechanics.

16. Why is the nervous system important for coordination and performance in sports?

The nervous system acts as the command center. It controls motor unit recruitment, balances muscle contraction/relaxation, regulates spatial awareness, and dictates reaction time.

17. Name any two lung volumes involved in breathing.

(Any two of the following are acceptable)

Tidal Volume (TV): Volume of air inspired or expired during a normal breath.

Inspiratory Reserve Volume (IRV): Extra volume of air that can be inspired forcefully after a normal tidal inspiration.

Expiratory Reserve Volume (ERV): Extra volume of air that can be expired forcefully after a normal tidal expiration.

Residual Volume (RV): Volume of air remaining in lungs after a maximal forced expiration.

18. How does biomechanical analysis improve performance in athletes?

It pinpoints technical flaws by measuring force, velocity, and angles.

Correcting these flaws enhances kinetic energy transfer, ensuring the athlete moves with optimal aerodynamic or mechanical efficiency.

19. What is an antagonist muscle and what role does it play in movement?

An antagonist is a muscle that opposes the movement of the primary mover (agonist).

Role: It relaxes or contracts eccentrically to control, slow down, or smoothly halt a movement, protecting joints from hyperextension.

Part C: Short Essay Questions

any 4 questions. Each carries 5 marks.

20. Explain the role of macronutrients (carbohydrates, proteins, fats) in supporting energy production during exercise.

Carbohydrates: The primary high-intensity fuel source. Stored as glycogen in muscles and liver; metabolized rapidly through anaerobic and aerobic pathways.

Fats: The primary fuel source for low-to-moderate intensity, long-duration exercise. Highly dense energy storage, but requires abundant oxygen to convert into usable ATP.

Proteins: Primarily used for structural tissue repair and enzyme synthesis. Contributes minimally to energy production (around 5% to 10%) unless carbohydrate stores are severely depleted.

21. How can understanding the major muscle groups and their functions help in applying correct biomechanics to prevent injuries during exercise?

Balanced Force Couples: Knowing muscle actions keeps opposing groups balanced (e.g., quadriceps vs. hamstrings), avoiding joint imbalances.

Kinetic Transfer: Helps verify that primary movers—like the gluteus maximus during squats—are doing the bulk of the work rather than overloading smaller accessory muscles.

Injury Prevention: Encourages proper form cues that prevent abnormal joint strain.

22. Explain how blood circulation changes during exercise to meet the increased oxygen and nutrient demands of muscles.

Cardiovascular Redistribution: Blood vessels feeding the digestive system and kidneys constrict (vasoconstriction), while vessels feeding working skeletal muscles dilate (vasodilation). This shunts up to 80-85% of cardiac output to active muscles.

Increased Output: Heart rate and stroke volume rise to pump oxygen-rich blood faster.

Venous Return: The skeletal muscle pump and respiratory pump speed up blood return to the heart.

23. How can understanding force, motion, and torque help athletes improve efficiency and prevent injuries during training or competition?

Force & Motion: Helps athletes maximize acceleration and momentum (e.g., optimizing foot strike force in sprinting).

Torque (Rotational Force): Vital for understanding rotational load on joints. Modifying leverage (moment arms) lowers stress on vulnerable areas like the lower back during heavy deadlifts.

24. How can understanding human motion help in enhancing sports performance and preventing injuries?

Movement Efficiency: Analyzing movement phases helps eliminate unnecessary actions, conserving energy.

Skill Mastery: Speeds up motor skill mastery by breaking sports actions into mechanical steps.

Load Distribution: Pinpoints overused segments to help balance stress across the entire kinetic chain.

25. How can understanding fuel utilization at different exercise intensities help athletes improve performance and recovery?

Intensity Mapping: Low intensity relies on fat oxidation; high intensity relies strictly on carbohydrates.

Pacing Strategies: Helps athletes manage glycogen stores to prevent "bonking" or hitting the wall during marathons.

Recovery Replenishment: Guides exact post-workout carbohydrate timing to restock glycogen and proteins to repair muscle fiber.

Part D: Essay Questions

any 2 questions. Each carries 15 marks.

26. Explain how posture and gait affect movement efficiency and the risk of musculoskeletal injuries. Provide examples from sports or daily activities.

Posture Basics: Discuss static vs. dynamic posture. Explain how neutral spine alignment evenly spreads compressive forces across intervertebral discs.

Gait Mechanics: Detail the phases of gait (Stance phase: heel strike, midstance, push-off; Swing phase).

Movement Efficiency: Poor posture or an altered gait (like overpronating your feet) creates a mechanical energy leak. The body has to exert extra muscular effort just to stabilize itself.

Injury Mechanisms: Chronic misalignment results in conditions such as patellofemoral pain syndrome, plantar fasciitis, or lower back strains.

Examples:

Sports: A runner with an overstriding gait experiences heavy braking forces, leading to shin splints.

Daily Life: Forward head posture from looking at phones alters thoracic curvature, causing neck and shoulder tension.

27. Describe how knowledge of energy systems can help athletes plan training, optimize performance, and delay fatigue during exercise.

The Three Key Energy Systems:

ATP-PCr System (Phosphagen): Anaerobic, ultra-short duration (0-10s), high power output (e.g., 100m sprint).

Glycolytic System (Lactic Acid): Anaerobic, short-to-medium duration (10s-2 mins), yields lactic acid/hydrogen ions as byproducts.

Oxidative System (Aerobic): Requires oxygen, long duration (over 2 mins), uses fats and carbohydrates, high total yield but slow delivery.

Training Design: Allows targeted conditioning. A weightlifter trains the ATP-PCr pathway with long recovery intervals, while an 800m runner develops anaerobic threshold buffering.

Optimizing Performance: Teaches athletes when to push and when to coast based on which fuel tank they are drawing from.

Delaying Fatigue: Aerobic training boosts mitochondrial density, helping clear lactic acid faster and pushing the anaerobic threshold further down the timeline.

28. Explain the three classes of levers in the human body with examples for each. Discuss how understanding lever mechanics helps in analyzing movement efficiency.

Lever Principles: Identify the components: Fulcrum (F) (joint), Effort (E) (muscle contraction force), and Resistance/Load (R) (weight of body segment or external object).

The Three Lever Classes:

First-Class Lever (E-F-R): The fulcrum sits in the middle. Example: Extension of the neck (Atlanto-occipital joint).

Second-Class Lever (F-R-E): The resistance sits in the middle. Offers high mechanical advantage (magnifies force). Example: Plantar flexion standing on tiptoes (metatarsophalangeal joint).

Third-Class Lever (F-E-R): The effort sits in the middle. Most common lever in the body. Sacrifices force production for speed and range of motion. Example: Biceps curl at the elbow joint.

Movement Efficiency Analysis:

Explains why shifting an external load closer to a joint (shortening the resistance arm) reduces the effort required by muscles.

Helps calculate the structural load placed on tendons, explaining why third-class setups demand massive internal muscular force to overcome small external loads.